

5 Find the particular solution of the differential equation

$$\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = 4\cos x,$$

given that, when $x = 0$, $y = -4$ and $\frac{dy}{dx} = 3$. [11]

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- 7 (a) Show that an appropriate integrating factor for

$$\sqrt{x^2-1} \frac{dy}{dx} + y = x^2 - x\sqrt{x^2-1}$$

is $x + \sqrt{x^2 - 1}$.

[4]

[illegible]

(b) Hence find the solution of the differential equation

$$\sqrt{x^2-1}\frac{dy}{dx}+y=x^2-x\sqrt{x^2-1}$$

for which $y = 1$ when $x = \frac{5}{4}$. Give your answer in the form $y = f(x)$. [7]

[illegible]